Edge cases (empty vs. missing) — Solution

> Back to the exercise · Hint 1 · Hint 2 · Solution

Where this exercise is saved: `practice/env_config_practice.py`, under its `# Edge cases` section — the same file also holds the Intermediate exercise, under an `# Intermediate` section. Run it with `cd practice && python env_config_practice.py`.

Builds on: your practice file of that same file — you are testing *that* loader against the two tricky `.env` states, so keep both sections in the one file. The `MissingConfigError` used below is the same class the Build Task ends up with in `practice/build_task/exceptions.py`.

Story — `env_config_practice.py` (Edge cases section): "missing" isn't one thing — `os.getenv()` returns `None` for a key that was never set, but an empty string `""` for one that's set to nothing, and a careless `if not value` check treats the two identically. This exercise decides, on purpose, which of those states a required secret should tolerate and which it must reject. **If not:** `require_env()` in the Build Task would ship with an untested assumption about empty strings, and a genuinely blank API key could sail through as if it were valid.

Solution

```
# practice/env_config_practice.py — Edge cases section
import os

class MissingConfigError(Exception):
    pass

def require_env(key: str) -> str:
    # why: strict on purpose — an API key that's set but empty is never
    # usable, so it's treated the same as not being set at all.
    value = os.getenv(key)
    if value is None or value == "":
        raise MissingConfigError(
            f"Required environment variable is missing: {key}"
        )
    return value

# case 1: key was never set at all
# how: pop(..., None) removes it if present, does nothing if not — never raises
os.environ.pop("OPENAI_API_KEY", None)
```

```
try:
    require_env("OPENAI_API_KEY")
except MissingConfigError as e:
    print(f"Case 1 - not set at all: {e}")

# case 2: key is set, but empty
# when: simulates a .env line like "OPENAI_API_KEY=" with nothing after the "="
os.environ["OPENAI_API_KEY"] = ""
try:
    require_env("OPENAI_API_KEY")
except MissingConfigError as e:
    print(f"Case 2 - set but empty: {e}")
```

Expected output:

```
Case 1 - not set at all: Required environment variable is missing:
OPENAI_API_KEY
Case 2 - set but empty: Required environment variable is missing: OPENAI_API_KEY
```